# 1 Introduction

## CONTROL SYSTEM

A group of physical components which provide a controlled output for a given input is known as control system.

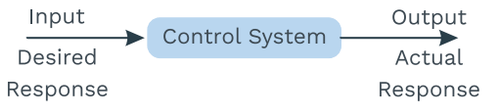



**Figure 1: Basic block diagram of control system**

The above figure represents a control system in its basic form, where a desired output is obtained for the specific input.

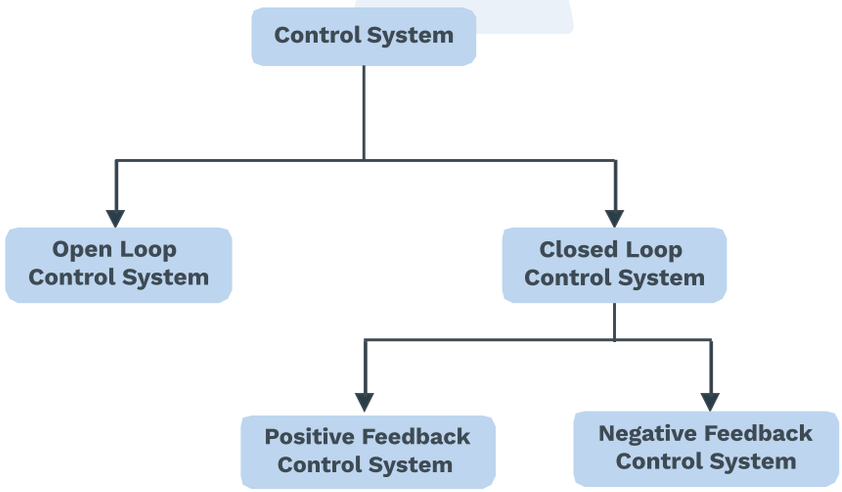
**Definitions**

**Control System**
It is a collection of components which are connected in such a manner that it gives the desired output.

## Types of Control System

Control system is of two types-

1. Open loop system
2. Closed loop system



### (a) Open loop control system

A system in which the output has no effect on the control action is called as open loop control system. In this system, output is not compared with the reference input as shown below.

**Gray Matter Alert!!!**

Positive feedback used for designing of oscillator and negative feedback is used for designing of an amplifier.

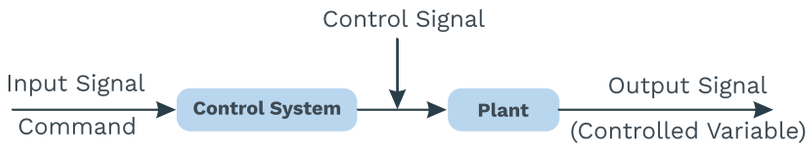



**Figure 2 : Representation of open loop control system**

The above block diagram represents an open loop control system where feedback element is absent.

The devices which are categorized under the open loop control system are-

1. Automatic washing machine
2. TV remote control
3. Volume of the audio system
4. Electric bulb
5. Electric hand drier

### Advantages of an open loop control system:

1. Low maintenance, therefore convenient
2. Inexpensive
3. Generally stable
4. Simple in construction and Design

### Disadvantages of an open loop control system:

1. Changes made in output cannot be corrected automatically.
2. They are inaccurate and therefore unreliable.
3. Effect of parameter variation is more.

**Rack your Brain**

Why performance analysis is not applicable to open loop control system?

### (b) Closed Loop Control System

It is a system in which output influence the control action with the help of feedback element, which converts the output into a suitable variable, which is then compared with the input signal.

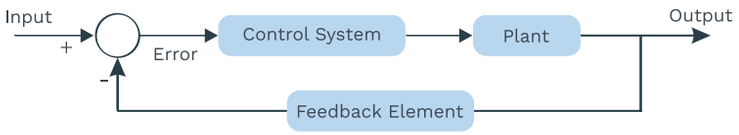



**Figure 3 : Basic block diagram of closed loop control system**

The above figure is the block diagram of closed loop system in which only feedback element is the additional feature as of open loop system.

### Some practical examples of closed loop control system

1. Cooling system in car
2. Servo voltage stabilizer
3. Water level controller
4. Automatic electric iron
5. Missile launched and auto tracking by radar.
6. Human respiratory system

### Advantages of closed loop system

1. Very precise and less prone to error.
2. High bandwidth (In negative feedback).
3. Errors can be corrected through the feedback signal.
4. Accurate and reliable.
5. Reduced effect of parameter variation.

### Disadvantages of the closed loop control system

1. Expensive
2. The designing of this system is complicated.
3. System may become unstable.
4. Gain reduces with negative feedback.

**Definitions**

In a **closed loop control system** the output has an effect on control action through a feedback element and hence it is also termed as feedback control system.

**Gray Matter Alert!!!**

The fundamental difference between an open loop control system and a closed loop control system is feedback element.

**Differences between an open loop and a closed loop control system:**

| Open Loop System | Closed Loop System |
|---|---|
| **1.** The system is simple and less expensive. | **1.** This system is costly because it is difficult to design, as the system consists of more number of components. |
| **2.** No Feedback element is used. | **2.** Feedback element is used. |
| **3.** 'Control Action' does not depends on the output signal. | **3.** 'Control Action' also depend on output signal. |
| **4.** Signal flows only from input to output. | **4.** Signal flows from an input to an output as well as from an output to an input. |
| **5.** Generally stable. | **5.** Stability depends on the system components. |

**Example 1:**
A good control system should be-
(a) sensitive to the parameter variation
(b) insensitive to the input command
(c) insensitive to the parameter variation but sensitive to input command.
(d) neither sensitive to the input command nor sensitive to the parameter variation.

**Solution: (c)**
Output should be very sensitive towards the input command but insensitive to the system parameter.

**Rack your Brain**



Why do we use negative feedback in control system?

**Example 2:**
Match List-I (Physical action or activity) with List-II (Category of system) and select the correct answer:

| List-I (Physical Action or Activity) | List-II (Category of System) |
|---|---|
| A. Human respiration system | **1.** Man-made control system |
| B. Pointing of an object with a finger | **2.** Natural including biological control system |
| C. A man driving a car | **3.** Control system whose components are both man-made and natural |
| D. A thermostatically controlled room heater | |

**Codes:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 2 | 3 | 1 |
| (b) | 3 | 1 | 2 | 1 |
| (c) | 3 | 2 | 2 | 3 |
| (d) | 2 | 1 | 3 | 3 |

**Solution:**(a)

**Note:**

- Performance analysis is not applicable to open loop control system because they are not subjected to any disturbances including input (which remains constant).
- An important advantage of feedback control system is that it reduces sensitivity to parameter variations.

**Rack your Brain**

Why do we preferred low pass filter in control system?

## LAPLACE TRANSFORM:

If f(t) is a function of time then its corresponding Laplace transform can be given as-

$$L\{f(t)\} = F(s)$$

Where, $F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}.dt$

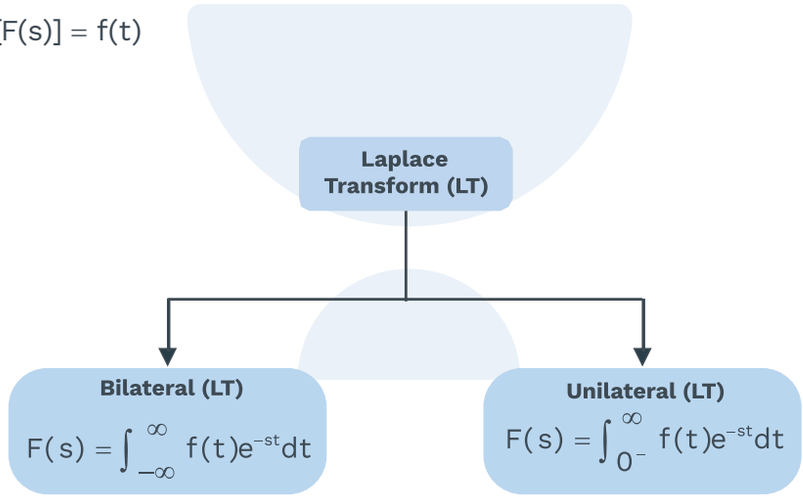
Here, $s = \sigma + j\omega$ is frequency variable.

We can also retrieve the original function f(t) by applying inverse Laplace transformation which is denoted as $\mathcal{L}^{-1}$.

$$\mathcal{L}^{-1}[F(s)] = f(t)$$

**Definitions**

**Laplace transform** is an integral transform that converts a function of real variable (time) to a function of complex variable (frequency).




**Note:**

- Lower limit is $0^-$ to include impulses.

### Properties of Unilateral transform:

1. If Laplace transform of x(t) is X(s), then

$$x(t) \xleftrightarrow{L.T.} X(s)$$

$$x(t \pm t_o) \xleftrightarrow{L.T.} e^{\pm st_o}X(s)$$

2. If

$$x(t) \xleftrightarrow{L.T.} X(s)$$

$$e^{\pm s_0 t}x(t) \xleftrightarrow{L.T.} X(s \mp s_0)$$

**Rack Your Brain**

What is the use of Laplace transform in control system?

**3.**

$$x(t) \xleftrightarrow{\text{L.T.}} X(s)$$

$$tx(t) \xleftrightarrow{\text{L.T.}} \frac{-dX(s)}{ds}$$

**4. Differentiation in Time:**

If $x(t) \xleftrightarrow{\text{L.T.}} X(s)$

Then $\frac{dx(t)}{dt} \xleftrightarrow{\text{L.T.}} sX(s) - x(0^+)$

$$\frac{d^2x(t)}{dt^2} \xleftrightarrow{\text{L.T.}} s^2X(s) - sx\left(0^+\right) - x'\left(0^+\right)$$

$$\frac{d^3x(t)}{dt^2} \xleftrightarrow{\text{L.T.}} s^3X(s) - s^2x\left(0^+\right) - sx'\left(0^+\right) - x''\left(0^+\right)$$

**5. Initial value theorem:**

$$\lim_{t\to 0^+} x(t) = \lim_{s\to\infty} sX(s)$$

Initial value theorem is valid only when x(t) is causal.

If $X(s) = \frac{N(s)}{D(s)}$

Then power of denominator D(s) is strictly greater than the power of numerator N(s).

**6. Final value theorem:**

$$\lim_{t\to\infty} x(t) = \lim_{s\to o} sX(s)$$

**Gray Matter Alert!!!**

Initial value theorem is valid only when x(t) is causal.

**Gray Matter Alert!!!**

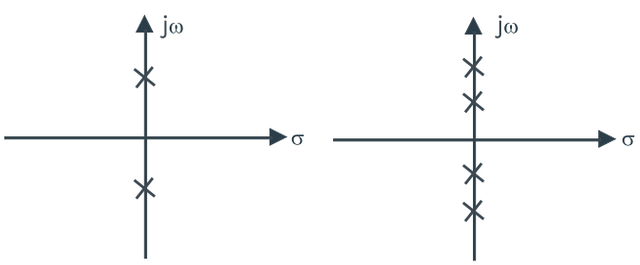
- Final value theorem is valid only when x(t) is causal.
- All the poles of X(s) should be strictly lie in left half of s-plane (LHP).

**Case-1.** If all the poles of transfer function X(s) are left hand pole

(a) Final value theorem (FVT) is applicable.

(b) Final value (F.V) is finite.

**Case-2.** If pole location of X(s) lies on jω axis.




**Non repeated poles**

Then, (a) Final value theorem is not applicable
(b) Final value is an indeterminate value.

**Example 3:**

If $L[f(t)] = \frac{\omega}{s^2 + \omega^2}$, then the value of $\lim_{t\to\infty} f(t)$

(a) is zero
(b) can not be determined
(c) is unity
(d) is infinite

**Solution:** (b)

$f(t) = \sin\omega t\ u(t) \xrightarrow{t=\infty}$ lies between $-1$ and $1$,

which is finite but cannot be determined.

**Previous Year's Questions**

The Laplace transform of i(t) is given by $I(s) = \frac{2}{s(1+s)}$ As $t \to \infty$, the value of i(t) tends to-
(a) 0
(b) 1
(c) 2
(d) $\infty$
Ans: (c)

## Unilateral Laplace Transform Formulae:

| S.No. | f(t) | F(s) = L[f(t)] |
|---|---|---|
| 1. | $\delta(t)$ | 1 |
| 2. | u(t) or 1 | $\frac{1}{s}$ |
| 3. | $e^{-at}$ or $e^{-at}\ u(t)$ | $\frac{1}{s+a}$ |
| 4. | $e^{at}$ | $\frac{1}{s-a}$ |
| 5. | t | $\frac{1}{s^2}$ |
| 6. | $\frac{t^2}{2}$ | $\frac{1}{s^3}$ |

| S.No. | f(t) | F(s) = L[f(t)] |
|---|---|---|
| 7. | $t^n$ | $\frac{n!}{s^{n+1}}$ |
| 8. | $te^{at}$ | $\frac{1}{(s-a)^2}$ |
| 9. | $t^n e^{-at}$ | $\frac{n!}{(s+a)^{n+1}}$ |
| 10. | $\sin\omega_0 t$ | $\frac{\omega_o}{s^2+\omega_o^2}$ |
| 11. | $\cos\omega_0 t$ | $\frac{s}{s^2+\omega_0^2}$ |
| 12. | $\sinh\omega_0 t$ | $\frac{\omega_o}{s^2-\omega_o^2}$ |
| 13. | $\cosh\omega_0 t$ | $\frac{s}{s^2-\omega_0^2}$ |

**Example 4:**

If the Laplace transform of signal y(t) is $Y(s) = \frac{1}{s(s-1)}$. Then its final value is-

(a) 0
(b) 1
(c) -1
(d) Unbounded

**Gray Matter Alert!!!**

- ULT x(t) = BLT [x(t) u(t)]

  $1 \xleftrightarrow{\text{ULT}} \frac{1}{s}$

  $u(t) \xleftrightarrow{\text{BLT}} \frac{1}{s}$

- In control system generally we use the unilateral laplace transform.

**Solution:** (d)

$$= \lim_{s \to 0} s \cdot \frac{1}{s(s-1)}$$

- Final value theorem is not applicable
- Final value $\to \infty$

1 pole line in RHP

$$\lim_{t \to \infty} y(t) = \lim_{s \to 0} sY(s)$$

So, final value is $\infty$, which is unbounded.

### Transfer function from differential equation:

Consider a differential equation as

$$f(t) = M.\frac{d^2x(t)}{dt^2} + f \cdot \frac{dx(t)}{dt} + Kx(t)$$

Where, Input = Force (F)
Output = Displacement (X)

$$TF = \frac{L[x(t)]}{L[f(t)]}\Bigg|_{\text{Initial Condition}=0}$$

$$TF = \frac{X(s)}{F(s)}$$

To convert the differential equation into s-domain we take the Laplace transform of the differential equation.

$$F(s) = M[s^2X(s) - sx(0) - x'(0)] + f[sX(s) - x(0)] + KX(s)$$

$\because$ All initial condition = 0

$\therefore$ $$F(s) = X(s)\,[Ms^2 + fs + k]$$

$$TF = \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + fs + K}$$

This is the transfer function of a mechanical translation system.

### Steps for determining the transfer function for a differential equation:

1. Take the Laplace transform of the differential equation and put all the initial conditions equals to zero.
2. If any constant term is present in differential equation, put it equal to zero because it represents the initial condition.
3. Specify the system output and the input.

**4.** Now take the ratio of the Laplace transform of the output and the Laplace transform of the input. The ratio of the Laplace transform of the output to the Laplace transform of the input will be the transfer function.

**Example 5:**

An LTI system is initially at rest when it is subjected to a unit step input, the response of the system is $y(t) = te^{-t}$, $t > 0$. Then the transfer function of the system is-

(a) $\frac{1}{(s+1)^2}$ (b) $\frac{1}{s(s+1)^2}$ (c) $\frac{s}{(s+1)^2}$ (d) $\frac{1}{s(s+1)}$

**Solution:** (c)

$y(t) = te^{-t}$, $t > 0$

$$Y(s) = L[y(t)] = \frac{1}{(s+1)^2}$$

$$X(s) = L[u(t)] = \frac{1}{s}$$

$$TF = \frac{Y(s)}{X(s)} = \frac{\frac{1}{(s+1)^2}}{\frac{1}{s}} = \frac{s}{(s+1)^2}$$

**Example 6:**

If the step response of a system is $c(t) = (1 - 10e^{-t})u(t)$, then obtain the transfer function of the system.

**Solution:**

$$c(t) = \left(1 - 10e^{-t}\right)u(t)$$

$$C(s) = L[c(t)] = \frac{1}{s} - \frac{10}{s+1}$$

Input = $R(t) = u(t)$

$$R(s) = \frac{1}{s}$$

$$TF = \frac{C(s)}{R(s)} = \frac{\frac{1}{s} - \frac{10}{s+1}}{\frac{1}{s}} = \frac{(s+1-10s)}{(s+1)} = \frac{1-9s}{s+1}$$

## PRACTICE QUESTIONS

**Q1** A control system is represented by a closed loop transfer function,

$$\frac{C(s)}{R(s)} = \frac{2(s-1)}{(s+2)(s+1)}$$

For a unit step input, the output is-

(a) $-3e^{-2t} + 4e^{-t} - 1$ (b) $-3e^{-2t} - 4e^{-t} + 1$

(c) Zero (d) Infinity

**Sol.** Correct answer is (a).

**Q2** A system is described by equation $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = x(t)$. Let x(t) be a rectangular pulse given by $x(t) = \begin{cases} 1, & 0 < t < 2 \\ 0, & \text{otherwise} \end{cases}$. Assuming that $y(0) = 0$ and $\frac{dy}{dt} = 0$ at $t = 0$, the Laplace transform of y(t) is-

**Sol.** $$Y(s) = \frac{\left(1 - e^{-2s}\right)}{s(s+2)(s+3)}$$

**Q3** In a closed loop control system -

(a) Control action is independent of output

(b) There is no feedback

(c) Control action is dependent on the output

(d) Output is independent of input

**Sol.** Correct answer is (c).

**Q4** By using feedback in control system, the sensitivity to parameter variation is improved. This is achieved at the cost of -
(a) Loss of system gain
(b) Stability
(c) Transient response
(d) Reliability

**Sol.** Correct answer is (a).

**Q5** The system which is capable of generating oscillations is-
(a) Open loop control system
(b) Closed loop control system
(c) Both (A) & (B)
(d) None of the above

**Sol.** Correct answer is (b).

**Q6** The transfer function of the system described by $\frac{d^2y}{dt^2}+\frac{dy}{dt}=\frac{du}{dt}+2u$ with u as input and y as output is -

(a) $\frac{s+2}{s^2+s}$ (b) $\frac{2}{s^2+s}$

(c) $\frac{2s}{\left(s^2+s\right)}$ (d) $\frac{s+1}{\left(s^2+s\right)}$

**Sol.** Correct answer is (a).

**Q7** A control system is defined by following mathematical relationship

$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 5x = 12\left(1 - e^{-2t}\right)$. The response of the system at $t \to \infty$ is -

(a) X = 2 (b) X = -2 (c) X = 6 (d) X = 2.4

**Sol.** Correct answer is (d).

**Q8** A function y(t) satisfies the following differential equation:

$$\frac{dy(t)}{dt} + y(t) = \delta(t)$$

Where $\delta(t)$ is the delta function. Assuming zero initial condition and denoting the unit step function by u(t), y(t) can be of the form -

(a) $e^{-t}u(t)$ (b) $e^{t}$ (c) $e^{t}u(t)$ (d) $e^{-t}$

**Sol.** Correct answer is (a).

**Q9** Which of the following device is/are example(s) of a closed loop control system?

(a) Washing machine
(b) Temperature controllers
(c) Human eye
(d) Bread toaster

**Sol.** Correct answers are (b) and (c).

**Q10** Actual input signal applied to a system in case of a closed loop system is-

(a) Controlling Signal (b) Actuating Signal

(c) An Error Signal (d) None of these

**Sol.** Correct answer is (b).

**Q11** If $L[f(t)] = \frac{2(s+1)}{s^2+2s+5}$ then $f(0^+)$ and $f(\infty)$ are given by -

(a) 2, 0 respectively (b) 0, 1 respectively

(c) 0, 2 respectively (d) $\frac{2}{5}$, 0 respectively

**Sol.** Correct answer is (a).

**Q12** If L[f(t)] = F(s), then f(t – T) is equal to -

(a) $e^{sT}F(s)$ (b) $e^{-sT}F(s)$ (c) $\frac{F(s)}{1+e^{sT}}$ (d) $\frac{F(s)}{1-e^{-sT}}$

**Sol.** Correct answer is (b).